

Exhibit 4

Dictionary of Computer Vision and Image Processing

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binocular: A system that has two cameras looking at the same scene simultaneously usually from a similar viewpoint. See also stereo vision.

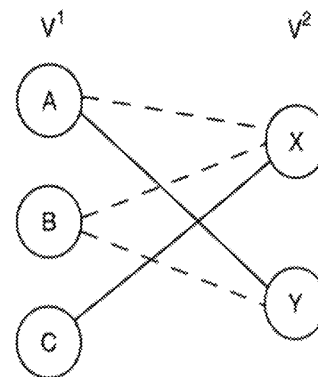
binocular stereo: A method of deriving depth information from a pair of calibrated cameras set at some distance apart and pointing in approximately the same direction. Depth information comes from the parallax between the two images and relies on being able to derive the same feature in both images.

binocular tracking: A method that tracks objects or features in 3D using binocular stereo.

biometrics: The science of discriminating individuals from accurate measurement of their physical features. Example biometric measurements are retinal lines, finger lengths, fingerprints, voice characteristics and facial features.

bipartite matching: Graph matching technique often applied in model based vision to match observations with models or stereo to solve the correspondence problem. Assume a set V of nodes partitioned into two non-intersecting subsets V^1 and V^2 . In other words, $V = V^1 \cup V^2$ and $V^1 \cap V^2 = \emptyset$. The only arcs E in the graph lie between the two subsets, i.e., $E \subset \{V^1 \times V^2\} \cup \{V^2 \times V^1\}$. This is the bipartite graph. The bipartite matching problem is to find a maximal matching

in the bipartite graph, in other words, a maximal set of nodes from the two subsets connected by arcs such that each node is connected by exactly one arc. One maximal matching in the graph below with sets $V^1 = \{A, B, C\}$ and $V^2 = \{X, Y\}$ pairs (A, Y) and (C, X) . The selected arcs are solid, and other arcs are dashed.



bit map: An image with one bit per pixel.

bit-plane encoding: An image compression technique where the image is broken into bit planes and run length coding is applied to each plane. To get the bit planes of an 8-bit gray scale image, the picture has a boolean AND operator applied with the binary value corresponding to the desired plane. For example, ANDing the image with 00010000 gives the fifth bit plane.

bitangent: See curve bitangent.

bitshift operator: The bitshift operator shifts the binary representation of each pixel to the left or right by a set number of bit positions. Shifting